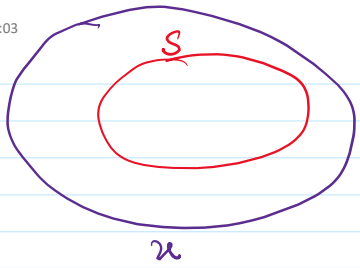
10) | Lecture 18 | Data Structure for Set Membership Query - Bloom Filter |

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Problem: Given a set S of universe u, $S \subseteq u$ $S = \{x_1 \dots x_n\}$ construct the data structure to ans queries of following form

Is $y \in S$? Yes / No

Data structure should be

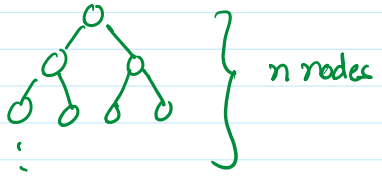
- Fast
- Space efficient

① Binary Search Tree

Input: A set $S \subseteq u$ s.t $|S| = n$

Space Complexity

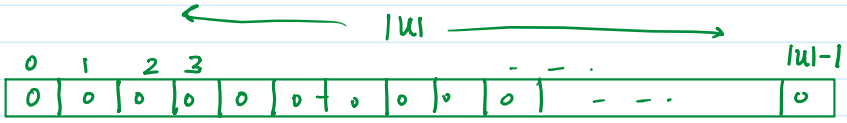
- Each node requires $O(\log |u|)$ bits
- Overall space complexity $O(n \log |u|)$



Search Time $O(\log n)$

② Bit Vector

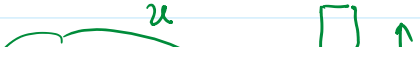
One bit for each possible element of universe.



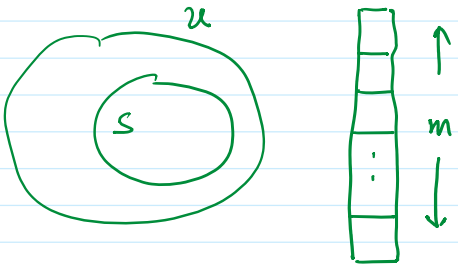
Space Complexity $O(|u|)$

Time Complexity $O(1)$

	Bit Vector	BST
Time Complexity	$O(1)$	$O(\log n)$
Space	$O(\|u\|)$	$O(n \log \|u\|)$



Space : | O(m) | O(n log |u|)



Universal Hashing : Definition

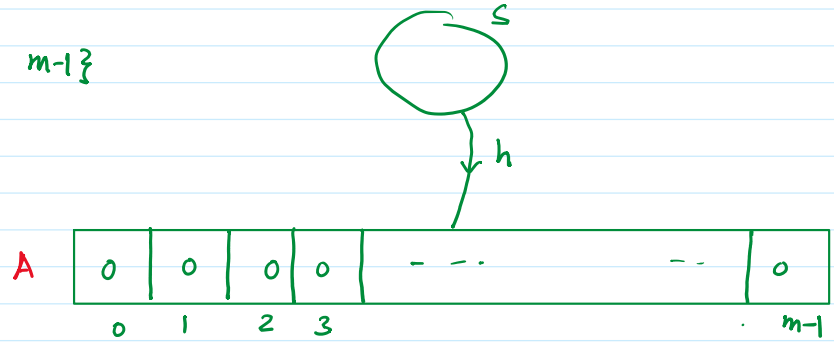
A family of hash fn $\mathcal{H}$.

- Choose a hash function $h \in \mathcal{H}$ uniformly at random
- We call $\mathcal{H}$ as universal hash family if $\forall$ pair of keys $k \neq k' \in S$

$$\Pr_{h \in \mathcal{H}} [h(k) = h(k')] \leq \frac{1}{m} \quad \text{--- (1)}$$

$h \in \mathcal{H}$ universal hash fn.

$h : S \longrightarrow \{0 \ldots m-1\}$



Bloom Filter (Burton Bloom 1970)

Step1 Design a family of Universal hash function $\mathcal{H}$

Choose k hash function $h_1, h_2 \ldots h_k \in_R \mathcal{H}$

Step2:

To insert: $x \in S$

$A[h_1(x)] \leftarrow 1$, $A[h_2(x)] \leftarrow 1$ - - $A[h_k(x)] \leftarrow 1$

To answer a query $y \in S$?

Check whether

- if $A[h_i(y)] = 1 \quad \forall \ i = 1 \ldots k$

if answer Yes Report $y \in S \quad \forall i \in [k]$

if answer No Report $y \notin S$

Ex $m = 10$ $K = 3$

$H(x) = \{ h_1(x), h_2(x), h_3(x) \}$

$m = 10$

0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0

Initialize with all 0's

Insert x_1

$H(x_1) = \{1, 4, 9\}$

0	1	2	3	4	5	6	7	8	9
0	1	0	0	1	0	0	0	0	1

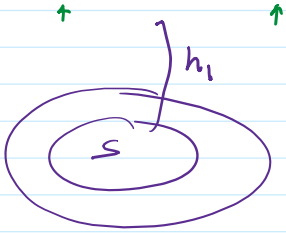
$h_1(x_1) = 1$, $h_2(x_1) = 4$, $h_3(x_1) = 9$

kn

Insert x_2

$H(x_2) = \{4, 5, 8\}$

0	1	2	3	4	5	6	7	8	9
0	1	0	0	1	1	0	0	1	1



Query Y_1

$H(Y_1) = \{2, 4, 8\}$ report NO.

Query Y_2

$H(Y_2) = \{1, 5, 8\}$ - report Yes (which might be wrong) False +ve

Given $|S| = n$, m , choose K

Want to choose K s.t false +ve prob is minimized.

How to compute false +ve prob

- Need to calculate the prob of "set of K- specific bits are 1"

After all elements of S have been hashed, what is the prob that a specific bit is still 0. $\left(1 - \frac{1}{m}\right)^{kn} \approx e^{-\frac{kn}{m}}$

Prob of false +ve : prob that a set of K specific bits are 1

$f := \left(1 - e^{-\frac{kn}{m}}\right)^K$

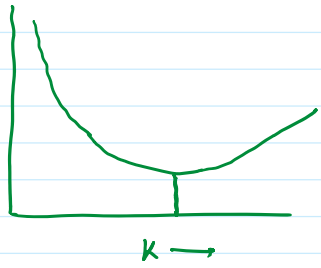
Prob of false +ve $f := \left(1 - \frac{1}{e^{\frac{kn}{m}}}\right)^K$

$\left(e^{-\frac{m}{n}}\right)$

< 1

increasing value of k - decrease value of f

increasing value of k - increase value of f



$$f = \left(1 - e^{-\frac{kn}{m}}\right)^k = \exp\left(\underbrace{k \ln\left(1 - e^{-kn/m}\right)}_{g}\right)$$

$$\frac{\partial g}{\partial k} = \ln\left(1 - e^{-kn/m}\right) + \frac{kn}{m} \frac{e^{-kn/m}}{1 - e^{-kn/m}}$$

$$\frac{\partial g}{\partial k} = 0 \quad \text{when} \quad k = \left(\frac{m}{n}\right) \ln 2$$

$$\frac{\partial^2 g}{\partial k^2} > 0 \quad \text{when} \quad k = \left(\frac{m}{n}\right) \ln 2$$

$$\boxed{k = \left(\frac{m}{n}\right) \ln 2}$$

Application of Bloom filter

1. Medium use bloom filter to avoid recommending article which a user has already read.
2. Web crawler used BF to identity if a URL is already downloaded.

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